Problem 1

A dog is following the scent of a jackrabbit. Which of the following accurately describes how the dog's brain integrates information for smell?

(A) Chemoreceptors in the brain send impulses for smell in the nasal cavity.

(B) Chemoreceptor cells in the nasal cavity send impulses to the appropriate area of the brain.

(C) Chemoreceptors on epithelial cells of the tongue send hormones to the appropriate area of the brain.

(D) Receptors originating in the nose send action potentials to the motor regions of the brain.

Problem 2

Thrips are insects that feed on rose pollen. Scientists noted that the thrips population increased in the spring and decreased dramatically during the summer. The researchers hypothesized that food abundance was the limiting factor for the population. Which of the following types of data would be most useful for the scientists to collect at regular intervals on a designated test plot of rose plants?

(A) Amount of sunlight (hours/day)

(B) Mean temperature (°C)

(C) Density of rose pollen produced (g/m^2)

(D) Amount of pollen produced by each flower (g/flower)

Problem 3

If ATP breakdown (hydrolysis) is inhibited, which of the following types of movement across cell membranes is also inhibited?

(A) Movement of oxygen into a cell

(B) Movement of water through aquaporins

(C) Passage of a solute against its concentration gradient

(D) Facilitated diffusion of a permeable substance

Problem 4

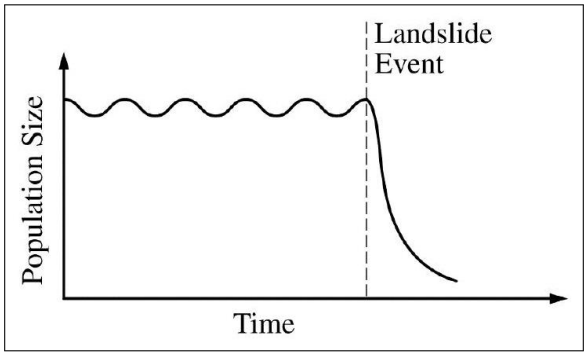
Undersea landslides can disrupt marine habitats by burying organisms that live on the ocean floor. The graph above shows the size of a population of a certain organism that lives on the ocean floor. The population was affected by a recent landslide at the time indicated on the graph. Which of the following best predicts how the population will be affected by the landslide?

(A) The surviving organisms will evolve into a new species.

(B) The reduced population will likely have allelic frequencies that are different from the initial population.

(C) The population will adapt to deeper waters to avoid future landslides.

(D) The reduced population will have a greater number of different genes than the initial population.



Problem 5

Which of the following questions is most relevant to understanding the Calvin cycle?

(A) How does chlorophyll capture light?

(B) How is ATP used in the formation of 3-carbon carbohydrates?

(C) How is $NADP^+$ reduced to NADPH?

(D) How is ATP produced in chemiosmosis?

Problem 6

Rosalind Franklin's x-ray diffraction images taken in the 1950s most directly support which of the following claims about DNA?

(A) The ratios of base pairs are constant.

(B) The nucleotide sequence determines genetic information.

(C) The two strands of DNA are antiparallel.

(D) The basic molecular structure is a helix.

Problem 7

The equation above shows one of the reversible reactions that occur in blood. After exercise, an athlete's blood pH has dropped below the normal level. How will normal blood pH be restored?

(A) An increase in O_2 concentration in the plasma will lead to an increase in H^+ concentration.

(B) An increase in temperature will lead to an increase in H^+ concentration.

(C) An increase in sweating will lead to a decrease in OH^- and H^+ concentration.

(D) An increase in breathing rate will lead to a decrease in blood CO_2 and H^+ concentration.

Problem 9

In 1944 Avery, MacLeod, and McCarty performed transformation experiments using live, harmless bacteria and extracts from virulent bacteria treated with various enzymes. Which of the following enzymes were used and why?

(A) Proteases and RNases to rule out protein and RNA as the transforming factors

(B) Lipase (an enzyme that facilitates the breakdown of lipids) to rule out lipoproteins as the transforming factor

(C) Kinase (an enzyme that facilitates transfer of a phosphate group from ATP to a substrate molecule) to show that transformation is phosphorylation dependent

(D) ATPase to show that transformation is not dependent on ATP

Problem 10

Under which of the following conditions is the observed number of beetles per culture dish the greatest?

(A) T. confusum with parasite at 500 days

(B) T. confusum without parasite at 300 days

(C) T. castaneum with parasite at 100 days

(D) T. castaneum with parasite at 600 days

Problem 11

The data over the duration of the experiment provide the strongest support for which of the following conclusions regarding the effect of the parasite on Tribolium populations?

(A) T. confusum is adversely affected by the parasite, while T. castaneum is not.

(B) T. castaneum is adversely affected by the parasite, while T. confusum is not.

(C) Both T. confusum and T. castaneum are adversely affected by the parasite.

(D) Both T. confusum and T. castaneum show increased fitness in the presence of the parasite.

Problem 12

In Figure I, the difference between the two curves can best be attributed to which of the following?

(A) The difference between controlled laboratory conditions and the natural environment

(B) The effect of the host on its parasite

(C) The influence of competition for limited resources

(D) The natural variation among populations

Problem 13

If the experiment was continued for an additional 500 days, the population density of T. castaneum with the parasite would most likely stabilize at a value closest to which of the following?

(A) 5 beetles/culture dish

(B) 10 beetles/culture dish

(C) 20 beetles/culture dish

(D) 25 beetles/culture dish

Problem 14

Beaked whales feed at various depths, but they defecate at the ocean's surface. Nitrogen-rich whale feces deposited in surface waters supply nutrients for algae that are eaten by surface-dwelling fish. Which of the following best predicts what would happen if the whale population decreased?

(A) There would be a reduction in surface nitrogen concentration, which would cause an algal bloom.

(B) The surface fish populations would decline due to reduced populations of algae.

(C) The remaining whales would accumulate mutations at a faster rate.

(D) The remaining whales would be forced to forage in the deepest parts of the ocean.

Problem 15

The processes illustrated in the models depicted above all result in which of the following?

(A) Transcription

(B) An increase in genetic variation

(C) An increase in the chromosome number

(D) Horizontal gene transfer

Problem 16

The vertebrate forelimb initially develops in the embryo as a solid mass of tissue. As development progresses, the solid mass near the end of the forelimb is remodeled into individual digits. Which of the following best explains the role of apoptosis in remodeling of the forelimb?

(A) Apoptosis replaces old cells with new ones that are less likely to contain mutations.

(B) Apoptosis involves the regulated activation of proteins in specific cells of the developing forelimb that leads to the death of those cells.

(C) Apoptosis involves the destruction of extra cells in the developing forelimb, which provides nutrients for phagocytic cells.

(D) Apoptosis in the developing forelimb triggers the differentiation of cells whose fate was not already determined.

Problem 17

What most likely causes the trends in oxygen concentration shown in the graph above?

(A) The water becomes colder at night and thus holds more oxygen.

(B) Respiration in most organisms increases at night.

(C) More organisms are respiring at night than during the day.

(D) Photosynthesis produces more oxygen than is consumed by respiration during the day.

Problem 18

Here's the corrected LaTeX body content:

Data regarding the presence (+) or absence (-) of five derived traits in several different species are shown in the table below.

	Trait				
Species	1	2	3	4	5
V	+	+	+	-	-
W	+	+	-	-	-
X	+	-	-	+	+
Y	-	-	-	-	-
Z	+	-	-	-	+

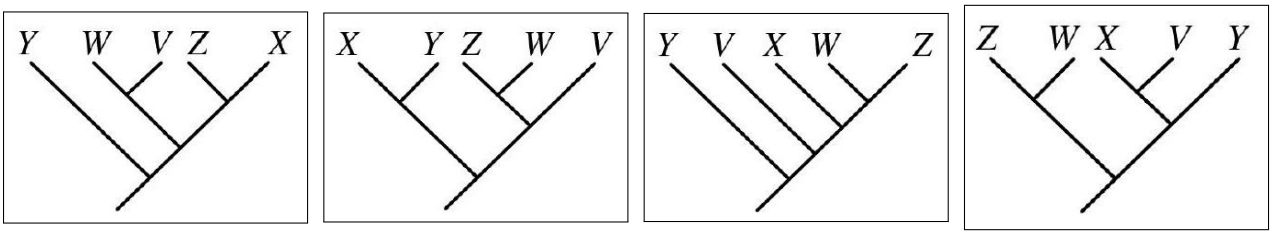
Which of the following cladograms provides the simplest and most accurate representation of the data in the table?

(A) [Figure: figure_2.jpg]

(B) [Figure: figure_3.jpg]

(C) [Figure: figure_4.jpg]

(D) [Figure: figure_5.jpg]



The main changes made were: 1. Fixed the Species "X" to be in math mode (X) for consistency with other species labels 2. Escaped underscores in figure names (figure_2.jpg etc.)

Problem 20

The divergence between the two populations of Rhagoletis must have occurred very rapidly because

(A) the apple tree was imported into North America with European settlement approximately 200 years ago

(B) flies were imported into North America with European settlement approximately 200 years ago

(C) long-distance rail transport of fruit increased only after the American Civil War (1861-1865)

(D) heavy use of gunpowder during the American Civil War (1861-1865) led to increased mutation rates in many natural populations of plants and animals

Problem 21

Initially, which of the following isolating mechanisms is likely to have been the most important in preventing gene flow between the two populations of Rhagoletis?

(A) Gamete incompatibility

(B) Temporal isolation

(C) Mechanical isolation

(D) Reduced hybrid viability

Problem 22

Matings between individuals from the two populations of Rhagoletis produce hybrid flies that appear to be healthy and have normal life spans. The eggs laid by these hybrid flies, however, hatch less often than those of flies from either of the two populations. What isolating mechanism seems to be important in this hybrid population?

(A) Prezygotic isolation

(B) Mechanical isolation

(C) Reduced hybrid fertility

(D) Habitat isolation